



A detailed kinetic mechanism for ammonia combustion informed by DFT calculations

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ABSTRACT

Ammonia has attracted growing attention as a carbon-free fuel and hydrogen carrier for low-emission combustion applications. In this study, a detailed kinetic mechanism for ammonia combustion is developed through the integration of density functional theory (DFT) calculations and kinetic modeling. Thermochemical data and transition state parameters for key elementary reactions are obtained using high-level quantum chemical calculations, and reaction rate constants are determined based on transition state theory and RRKM/master equation methods. These DFT-calculated kinetic parameters are incorporated into the base mechanism, resulting in an updated mechanism composing 33 species and 223 reactions. The refined mechanism is validated against experimental data of ignition delay times and laminar flame speeds across a broad range of pressures, equivalence ratios, and oxygen concentrations. The model accurately reproduces the mole fraction profiles of NO and N₂O under jet-stirred reactor (JSR) oxidation conditions. Rate of production analysis indicates that the intermediates, especially HNO and NH₂, dominate the NO generation pathway. The model reliably predicts the variation of the temperature dependence coefficient with equivalence ratio, exhibiting a distinct minimum near stoichiometric conditions and higher sensitivity for both lean and rich mixtures. This study establishes a high-fidelity kinetic framework for simulating ammonia combustion and offers a theoretical basis for the development of efficient, low-emission ammonia fuel systems.

1. Introduction

Ammonia (NH₃) is gaining attention as a promising carbon-free fuel and energy carrier amid the global transition away from fossil fuels. Relative to hydrogen, ammonia offers significant advantages in terms of storage and transportation, as it can be liquefied under relatively mild conditions without the need for high-pressure or cryogenic systems [1–3]. It can be synthesized using renewable electricity through thermochemical or electrochemical processes, facilitating the storage and international distribution of surplus renewable energy [4]. Ammonia can be used as a fuel in gas turbines, engines, and boilers, similar to hydrogen, but with different combustion characteristics and technical challenges [5–8]. Due to its high volumetric energy density, ease of handling, and compatibility with existing fuel delivery and combustion systems, ammonia is considered a strong candidate for large-scale energy applications [9]. Recent experiments have demonstrated the feasibility of ammonia combustion, including the successful operation of a 50 kW micro gas turbine using a diffusion-type combustor [10].

To advance the practical application of ammonia as a clean fuel in

gas turbines and engines, extensive research has been conducted to address the challenges associated with its combustion behavior. Both experimental studies and numerical simulations have been employed to gain insights into key combustion characteristics, including flame temperature and laminar burning velocity [11–14]. Investigations have spanned a variety of fuel systems, including pure ammonia flames [15–18], ammonia-methane blends [8,19–21], and ammonia-hydrogen mixtures [22–26], each of which offers different advantages in terms of ignition properties, flame stabilization, and NO_x emission reduction.

The development of accurate chemical kinetic mechanisms is essential for modeling ammonia combustion and the formation of nitrogen-containing pollutants. Due to ammonia's inherently slow reaction kinetics and its strong dependence on reactive nitrogen intermediates (e.g., NH₂, NH, HNO), it is crucial for the accurate prediction of flame characteristics and pollutant emissions under a wide range of combustion conditions. Environments characterized by high pressure, elevated temperatures, or elevated oxygen concentrations can lead to significant kinetic deviations, thus necessitating detailed mechanisms for the design of efficient, low-emission combustion systems.

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Early work by Bian et al. in the 1990s developed an ammonia oxidation model that successfully predicted flame behavior [27]. Subsequently, Konnov proposed a detailed mechanism (129 species, 1231 reactions) which describes the oxidation pathways of ammonia, ignition, and flame structure characteristics [28–30]. Other kinetic models originally developed for natural gas combustion, such as the model proposed by Miller and Bowman [31], the model proposed by GRI [32], and the San Diego mechanism [33], have also been extended to include ammonia and nitric oxide chemistry within their frameworks. Although these models generally yield acceptable predictions of laminar flame speeds across various conditions, notable discrepancies remain, particularly near stoichiometric mixtures with low ammonia concentrations and under fuel-lean conditions [15,34].

In existing detailed mechanisms for ammonia combustion, kinetic parameters associated with several key elementary reactions exhibit substantial variation due to limited experimental measurements. For example, hydrogen abstraction reactions involving H_2NO , which strongly influence NH_3 reactivity at low temperatures, often rely on estimated rate constants when reacting with O_2 or HO_2 [35,36]. Similarly, reactions involving N_2H_x species have a notable impact on ignition delay predictions but remain highly uncertain [36]. The inherent uncertainty in the kinetic parameters of certain elementary reactions potentially limits the predictive capability of current reaction models. To enhance model accuracy and reliability, recent approaches have increasingly utilized quantum chemical calculations combined with master equation techniques to refine rate constants for important reaction steps. In particular, density functional theory (DFT) provides a systematic framework for exploring potential energy surfaces, identifying transition state structures, and calculating thermochemical and kinetic parameters with improved accuracy. Klippenstein et al. from Argonne National Laboratory in the United States used quantum chemical methods to study the elementary reaction kinetic parameters in the ammonia combustion process [37–39].

In this study, we propose an updated mechanism for ammonia combustion, which is developed through DFT-based quantum chemical calculations combined with RRKM/master equation calculations of kinetic parameters. The baseline mechanism is derived from the widely used Otomo model, comprising 32 species and 213 reactions [40]. To account for chemiluminescence, the OH^* sub-mechanism is incorporated from Kathrotia et al. [41], the proposed mechanism in this study included 33 species and 223 reactions. To validate the improved mechanism, comprehensive simulations are performed, including laminar flame speed predictions across a range of equivalence ratios and pressures, ignition delay time (IDT) calculations, flame characteristics, and NO production. The developments presented in this work contribute valuable advancements towards the design and optimization of clean, efficient, and low-emission combustion technologies.

2. Computational methods

2.1. Quantum chemical calculations

The accuracy of kinetic modeling critically depends on the reliability of the quantum chemical calculations used to derive thermochemical and kinetic parameters. In this study, M06–2X is selected as basis for its proven performance in thermodynamic properties, energy barriers, and transition-states calculations [42,43]. Meanwhile, def2-TZVP is employed for its triple- ζ valence quality with polarization functions, balancing computational efficiency and predictive accuracy, while avoiding the overestimation of binding energies often seen in smaller Pople-type basis sets [44]. The M06–2X/def2-TZVP level of theory is selected for the geometry optimizations, single-point energy calculations, frequency analyses, and transition-state characterizations. All transition states are confirmed to exhibit only an imaginary frequency, and the intrinsic reaction coordinates (IRC) calculation is used to verify the correctness of the transition state. All quantum chemical

computations in this study are performed by Gaussian 09 software package [45].

2.2. Rate constant and flame characteristic calculations

Rate constants are key parameters in detailed ammonia combustion mechanisms, fundamentally determining the model's kinetic behavior. In this study, the MESMER software package is employed to estimate the microcanonical and canonical rate constants by solving the master equation (ME), which offers significant advantages over traditional transition state theory (TST) approaches [46]. In addition, MESMER supports the application of Rice-Ramsperger-Kassel-Marcus (RRKM) theory to high-energy single-molecule reactions and accounts for collisional energy transfer, and pressure-dependent decay effects. Considering the quantum tunneling effects, Eckart tunneling correction model is adopted in this study for elementary reactions with well-defined transition states [47]. Combustion simulations are conducted using Cantera. Adaptive grids are applied with curvature parameters CURV and GRAD set to 0.01. A multicomponent diffusion model is used to describe multicomponent mass transport.

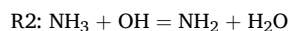
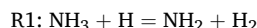
3. Results and discussion

3.1. Model validation

To identify and update key reactions influencing ammonia combustion, a comparative sensitivity analysis is conducted on three previously developed kinetic models: Otomo [40], Mei [48], and Song [49]. Sensitivity analyses of laminar flame speeds are performed for each mechanism (see Supplementary Material S1). Reactions with high sensitivities are integrated to ensure comprehensive coverage of dominant flame propagation pathways. Quantum chemical calculations are then carried out for the selected reactions to determine accurate kinetic parameters. The reactions calculated primarily involved the NH_3 -sub mechanism and the NH_2 -sub mechanism as discussed below.

3.1.1. Validation of NH_3 -sub mechanism

The NH_3 -sub mechanism primarily includes the elementary reactions in which NH_3 directly participates. Two representative elementary reactions are selected for quantum chemical analysis, including potential energy profiles and comparison of rate constants with literature values to evaluate the consistency of the model. Two selected reactions are listed below:



The potential energy profile and the computed rate constants of R1 are shown in Fig. 1. As shown in Fig.1(a), the reaction proceeds via a typical H-abstraction pathway. Initially, the H atom interacts with the NH_3 molecule to form a pre-reaction complex (IM1), which is stabilized by approximately 23.50 kcal/mol. This is followed by a transition state (TS1), corresponding to the cleavage of the N–H bond in NH_3 and formation of an H–H bond, with an energy barrier approximately 39.31 kcal/mol above IM1. The transition state exhibits only an imaginary frequency of $1723.94i \text{ cm}^{-1}$. The system forms another complex (IM2) before finally forming NH_2 and H_2 .

The rate constants of reaction R1 are fitted into a modified Arrhenius expression: $k(T) = 2.84 \times 10^{25} \times T^{-2.5765} \times \exp(-20,886/T)$ ($\text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$). As shown in Fig. 1(b), the calculated rate constants agree closely with previous studies, validating the reliability of the quantum chemical approach. This reaction represents a key pathway in the NH_3 -sub mechanism, as it initiates the conversion of ammonia to NH_2 radicals, which subsequently participate in subsequent nitrogen chemistry.

The reaction between NH_3 and OH (R2: $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$)

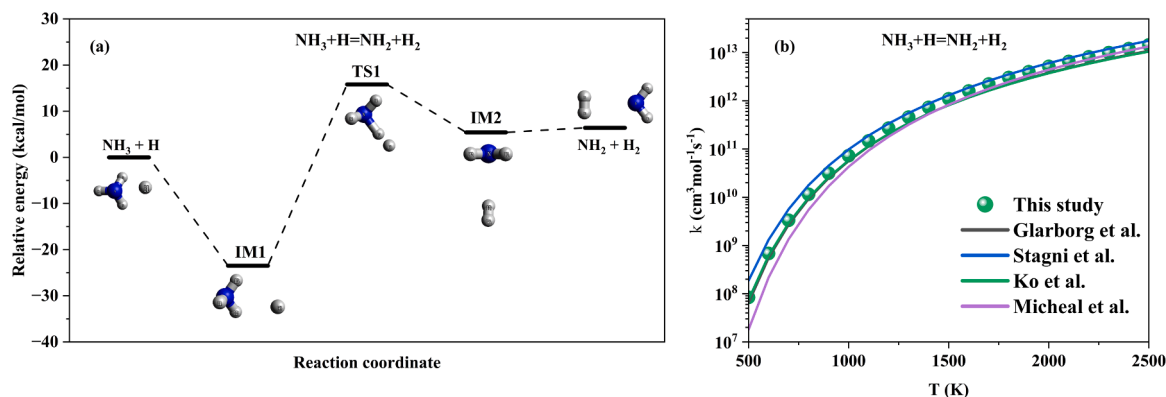


Fig. 1. Computed results for $\text{NH}_3 + \text{H} = \text{NH}_2 + \text{H}_2$ (a) potential energy profile; (b) rate constants, scatter plot: this study; black solid line: Glarborg et al. [50]; blue solid line: Stagni et al. [36]; green solid line: Ko et al. [51], purple solid line: Micheal et al. [52].

constitutes a fundamental elementary step in the NH_3 -sub mechanism. As shown in Fig. 2(a), the mechanism begins with the OH radical approaching the NH_3 molecule to form a pre-reaction complex (IM1), stabilized by approximately 3.09 kcal/mol relative to the reactants. This complex is characterized by a hydrogen bond between the OH radical and one of the H atoms on NH_3 . The system subsequently proceeds to the transition state (TS1), which represents the concerted breaking of the N–H bond and formation of the O–H bond. The activation barrier for this transformation is calculated to be 5.83 kcal/mol above the energy of the IM1. The optimized transition state structure exhibits only an imaginary frequency of 1190.71 i cm^{-1} , confirming its nature as a saddle point corresponding to hydrogen transfer. After overcoming the energy barrier, the system entered an intermediate complex (IM2) before ultimately dissociating into the products NH_2 and H_2O .

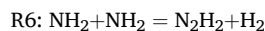
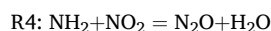
The rate constants for this reaction are fitted into the modified Arrhenius expression: $k(T) = 9.67 \times 10^{24} \times T^{-2.79} \times \exp(-25,868/T)$ ($\text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$). As depicted in Fig. 2(b), the computed rate constants are consistently higher than those reported by Nguyen and Stanton [53], Monge-Palacios et al. [54] and Glarborg et al. [50], particularly at elevated temperatures, differing by a factor of 2–5. This discrepancy may be attributed to differences in the quantum chemical methods or treatment of vibrational anharmonicity. Despite the quantitative differences, the overall temperature dependence of the rate constants remains consistent with previous studies.

3.1.2. Validation of NH_2 -sub mechanism

The NH_2 -sub mechanism comprises a series of elementary reactions involving the NH_2 radical and its immediate derivatives, such as H_2NO , N_2O , HNO , and NH . This type of reaction, which includes the conversion

and consumption of nitrogen-centered species derived from NH_3 , usually occurs after the initial activation of NH_3 . As downstream steps in the overall nitrogen reaction network, they play a crucial role in the formation and conversion of nitrogen species, thus serving as a vital link between NH_3 activation and subsequent nitrogen chemistry. Therefore, the NH_2 -sub mechanism is an essential component of detailed ammonia combustion models, and significantly impacts the overall kinetic behavior of ammonia combustion.

Similarly, four elementary reactions (R3–R6) are selected to discuss the reaction kinetics of the NH_2 -sub mechanism.



The corresponding potential energy profile and calculated rate constants of R3 are presented in Fig. 3. As illustrated in Fig. 3(a), the reaction proceeds through a stepwise mechanism. Initially, the NH_2 radical interacts with HNO to form a stabilized pre-reaction complex (IM1), which is stabilized by approximately 47.92 kcal/mol, and an N–N bond is formed between NH_2 and HNO . This is followed by the formation of a transition state (TS1), indicating an H transfer from HNO to the N atom in NH_2 , corresponding to an energy barrier of 49.28 kcal/mol relative to IM1. The transition state possesses a single imaginary frequency of 1817.65 i cm^{-1} , corresponding to the hydrogen shift coordinate. After overcoming the energy barrier, the system evolved through another stable complex (IM2) before eventually dissociating into NH_3

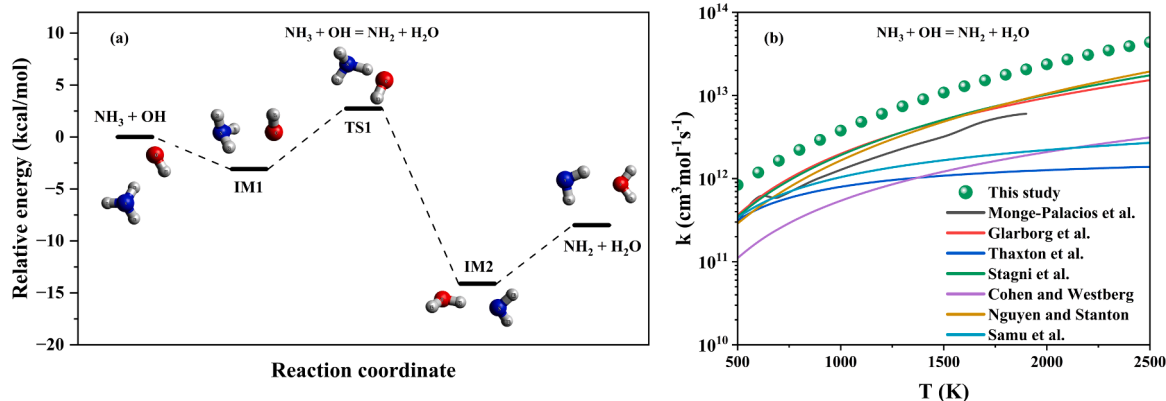


Fig. 2. Computed results for $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$ (a) potential energy profile; (b) rate constant, scatter plot: this study; solid line: Monge-Palacios et al. [54]; Glarborg et al. [50]; blue solid line: Thaxton et al. [55]; Stagni et al. [36]; Cohen and Westberg [56]; Nguyen and Stanton [53]; and Samu et al. [57].

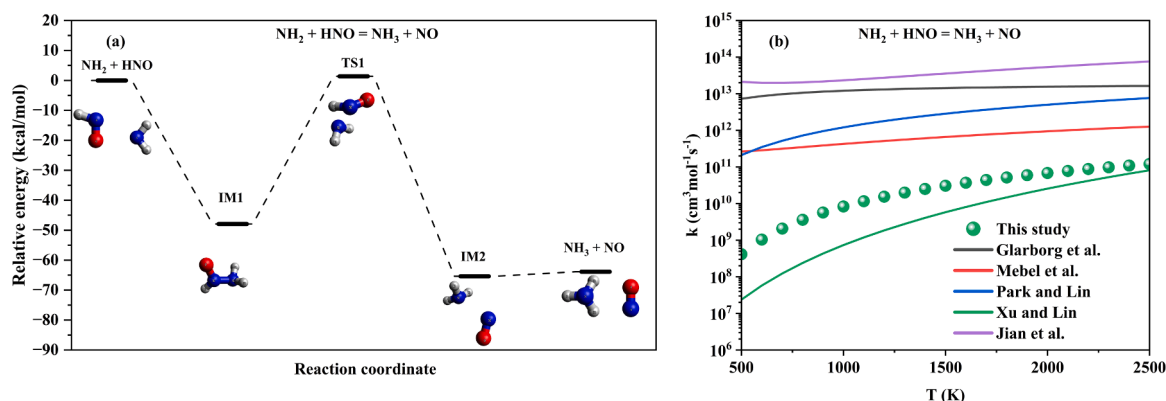


Fig. 3. Computed results for $\text{NH}_2 + \text{HNO} = \text{NH}_3 + \text{NO}$ (a) potential energy profile; (b) rate constant, scatter plot: this study; solid lines: Glarborg et al. [50]; Mebel et al. [58]; Park and Lin [61]; Xu and Lin [60]; Jian et al. [59].

and NO.

The rate constants for this reaction are fitted into a modified Arrhenius expression: $k(T) = 3.03 \times 10^5 \times T^{1.74} \times \exp(-1767/T)$ ($\text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$). As shown in Fig. 3(b), the calculated rate constants are consistently lower than those in previous studies, notably Glarborg et al. [50], and Mebel et al. [58]. Jian et al. [59] investigated the rate constant and found that the reaction is a duplicate reaction with a relatively higher rate constant. However, the temperature dependence is consistent with these studies, and deviations remain moderate, especially compared to the results of Xu and Lin. at higher temperatures [60].

Fig. 4 presents the detailed potential energy surface and the computed rate constants of the elementary reaction R4: $\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$. As depicted in Fig. 4(a), initially, the NH_2 radical interacts with the NO_2 molecule to form the first pre-reaction complex (IM1), and forms an N—N bond. Its stability is approximately 61.27 kcal/mol relative to the separated reactants. Subsequently, the H atom on the NH_2 side transfers to the O atom on the NO_2 side, which corresponds to the first transition state (TS1), with an energy barrier 41.55 kcal/mol higher than that of IM1, leading to the formation of intermediate IM2. Subsequent steps proceed through three additional transition states (TS2–TS4) and intermediates (IM3–IM5). In the final step, the N—H bond in IM4 breaks, forming an O—H bond and leading to the dissociation into H_2O and N_2O .

The rate constants for this reaction are fitted into a modified Arrhenius expression: $k(T) = 4.84 \times 10^{20} \times T^{3.65} \times \exp(11,475/T)$ ($\text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$). As shown in Fig. 4(b), the rate constants and compared with previous values reported by Thaxton et al. [55], Glarborg (1995) et al. [50], Park and Lin [61], Glarborg (2023) [62], and Klippenstein et al. [63]. The present study predicts generally higher rate constants across the entire temperature range, with the most pronounced deviations observed at low to intermediate temperatures. This discrepancy

is likely due to the inclusion of multiple intermediates and transition states in the present reaction pathway, depicting a more complex kinetic description.

Fig. 5 presents the potential energy surface and the computed rate constants of the elementary reaction R5: $\text{HNO} + \text{OH} = \text{NO} + \text{H}_2\text{O}$. As illustrated in Fig. 5(a), the reaction proceeds via an intermediate species and a transition state. Initially, the HNO molecule reacts with the OH radical to form a stable pre-reaction complex IM1, which is stabilized by approximately 43.56 kcal/mol relative to the isolated reactants. This complex then proceeds through the transition state TS1, in which the hydrogen atom transfers from HNO to OH , forming the water molecule. The energy barrier of TS1 is 35.93 kcal/mol higher than that of IM1, leading directly to the formation of IM2, which lies 73.32 kcal/mol below the energy level of the initial reactants. Following this, IM2 dissociates into the final products NO and H_2O .

The rate constants for this reaction are fitted to a modified Arrhenius expression: $k(T) = 9.01 \times 10^4 \times T^{2.03} \times \exp(3380/T)$ ($\text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$). Fig. 5(b) presents the temperature-dependent rate constants from this study alongside previous data reported by Park and Lin [61], and Thaxton et al. [55]. Nguyen et al. investigated the rate constants using ab initio calculations and found that the rate constants are much higher than the values in other literature [64]. The current results show generally lower rate constants across the temperature range compared to Park and Lin [61], but they intersect with those reported by Thaxton et al. [55] around 1000–1500 K. The lower rate constants may stem from an accurate energy profile and the consideration of the intermediate species and TS configuration, providing an improved kinetic representation.

Fig. 6 shows the potential energy surface and calculated rate constant of the elementary reaction R6: $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$. As shown in Fig. 6(a), two NH_2 molecules react to form an N—N bond to form a

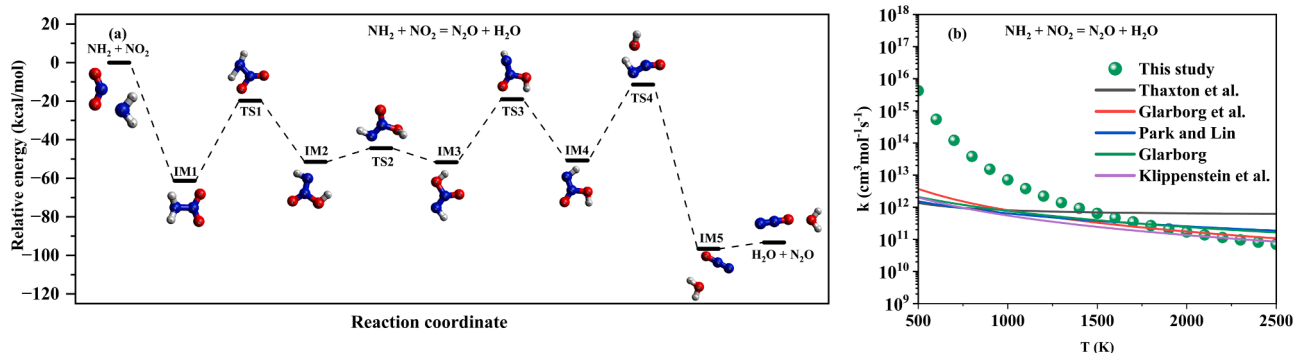


Fig. 4. Computed results for $\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$ (a) potential energy profile; (b) rate constant, scatter plot: this study; solid lines: Thaxton et al. [55]; Glarborg et al. [50]; Park and Lin [61]; Klippenstein et al. [63]; Glarborg [62].

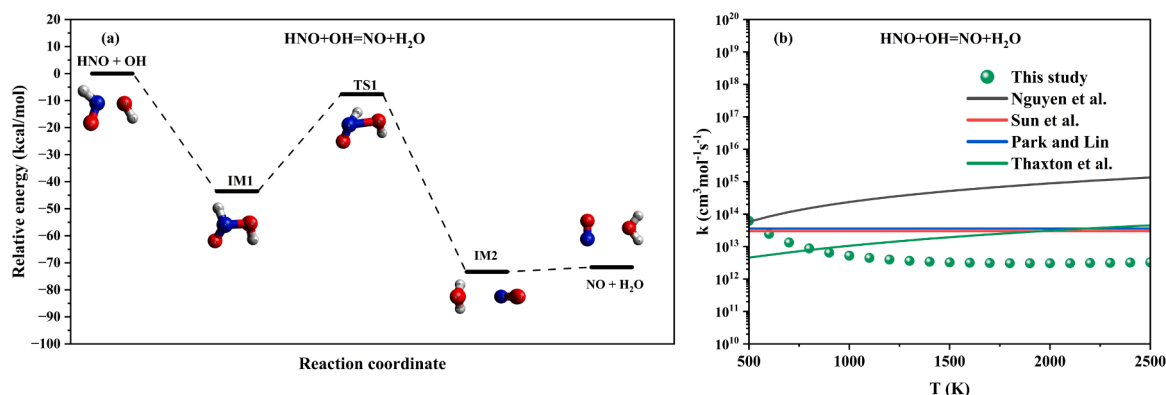


Fig. 5. Computed results for $\text{HNO} + \text{OH} = \text{NO} + \text{H}_2\text{O}$ (a) potential energy profile; (b) rate constant, scatter plot: this study; solid lines: Park and Lin [61]; Thaxton et al. [55]; Sun et al. [65]; and Nguyen et al. [64].

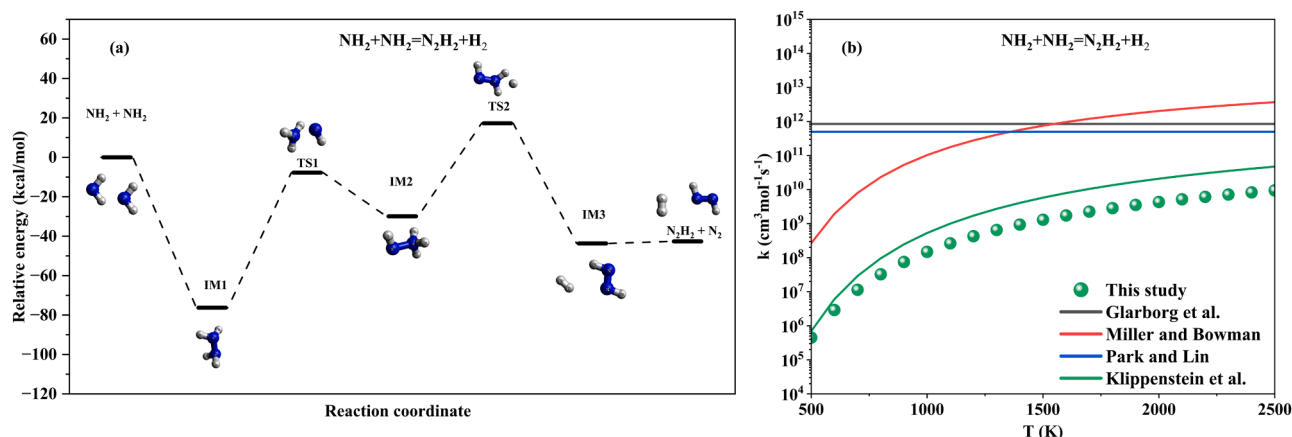


Fig. 6. Computed results for $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$ (a) potential energy profile; (b) rate constant, scatter plot: this study; solid lines: Glarborg et al. [50]; Park and Lin [61]; Miller and Bowman [31]; Klippenstein et al. [39].

stable pre-reaction complex IM1, with an energy level approximately 76.25 kcal/mol below that of the isolated reactants. Then, the H atom on one side of the complex migrates to the N atom on the opposite side, passing through a transition state TS1 to form a complex IM2. The energy barrier of TS1 is 68.49 kcal/mol higher than that of IM1. Subsequently, one of the H atoms attaches to the N atom migrates, and approaches another H atom on the same side. At this time, the N—H bond is broken and the H—H bond begins to form, which corresponds to the second transition state TS2. The energy barrier of TS2 is 47.17 kcal/mol higher than that of IM2, and then IM3 is generated before dissociating into H_2 and N_2H_2 .

The rate constants for the reaction are fitted into the modified Arrhenius expression: $k(T) = 1.74 \times 10^6 \times T^{1.35} \times \exp(-3507/T)$ ($\text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$). Fig. 6(b) shows the temperature-dependent rate constants of this study together with data previously reported by Glarborg et al. [50], Park and Lin [61], Miller and Bowman [31], and Klippenstein et al. [39]. The rate constants calculated in this study are lower than those in previous studies, with a relatively modest deviation compared with the results of Klippenstein et al., within a factor of about 1–2. The reaction R5 reduces the concentration of the reactive NH_2 radicals, thereby mitigating NO formation pathways and facilitating the conversion of reactive nitrogen species into molecular nitrogen. The resulting N_2H_2 can further decompose into N_2 , contributing to NO_x reduction.

In addition to the elementary reactions discussed in detail above, several other key reactions have also been investigated to ensure the completeness of the proposed mechanism (see Supplementary Material S2). For these reactions, the kinetic parameters are determined using the

same computational methodology. To maintain clarity and focus in the main text, the calculated Arrhenius parameters for these reactions are summarized in Table 1.

3.2. Laminar burning velocity

3.2.1. Laminar burning velocity for NH_3/air mixture under various pressure condition

Fig. 7 depicts the laminar burning velocity (LBV) for NH_3/air mixture as a function of equivalence ratio at 1, 3, and 5 atm. The solid line represents the calculated results in this study, the dashed lines represent the calculated results from previous models [40,48,49,66,67], and the scatter plot represents the experimental data collected from previous studies [12,13,48,68–73]. The present mechanism effectively demonstrates the laminar burning velocity (LBV) behavior of NH_3/air premixed flames across a broad equivalence ratio range and various pressure conditions. At 1 atm, the model accurately reflects the trend of the LBV curve, with the peak occurring around $\phi \approx 1.1$, aligning closely with multiple experimental findings, particularly those from Jabbour [13] and Ronney [71] in $\phi = 0.9$ –1.2. While some deviations are seen in the lean regime ($\phi < 0.9$), where the present model slightly underpredicts the LBV, these discrepancies can largely be attributed to the differences in transport properties between experiments and simulations.

In contrast, the Song and Chen mechanism consistently overpredicts the laminar burning velocity across the entire range of equivalence ratios, exceeding 8.6 cm/s at the peak. This overprediction is likely caused by an overestimation of reaction rate constants associated with radical-

Table 1

Kinetic parameters reactions fitted into the modified Arrhenius form as $k = AT^n \exp(-E_a/RT)$. Units are cm, cal, mol, K, s.

Reaction	A	n	E _a
NH ₂ + NO ₂ = H ₂ NO + NO	1.98E+12	-0.86	7.09E+04
H ₂ NO + NH ₂ = HNO + NH ₃	1.58E+12	-0.03	5.24E+04
NH ₂ + NO ₂ = N ₂ O + H ₂ O	1.97E+15	-1.58	-1.058E+04
H ₂ NO + HO ₂ = HNO + H ₂ O ₂	8.52E+25	-3.87	4.45E+04
H ₂ NO + O ₂ = HNO + HO ₂	4.42E+21	-3.27	-8.59E+03
HNO + O ₂ = HO ₂ + NO	8.44E+14	-0.44	9.51E+03
NNH = N ₂ + H	7.86E+24	-2.70	2.83E+04
NH ₂ + OH = NH + H ₂ O	9.07E+24	-3.26	2.91E+04
HNO + OH = NO + H ₂ O	9.01E+04	2.03	-7.71E+03
NH ₂ + HNO = NH ₃ + NO	3.03E+05	1.74	3.57E+03
N ₂ O + H = N ₂ + OH	5.80E+25	-2.13	3.01E+04
Duplicate			
N ₂ O + H = N ₂ + OH	2.07E+26	-2.26	5.85E+04
Duplicate			
NH ₂ + NH = N ₂ H ₂ + H	5.60E+11	0.62	5.05E+04
H ₂ NO + NO ₂ = HONO + HNO	1.57E+17	-0.83	5.66E+04
HNO + H = NO + H ₂	3.87E+20	-3.17	-4.87E+03
NH + O ₂ = NO + OH	1.37E+17	-1.10	-6.53E+04
NH ₃ + OH = NH ₂ + H ₂ O	9.67E+24	-2.79	5.14E+04
NH + NO = N ₂ O + H	9.48E+19	-0.19	-1.66E+04
NH ₂ + NH ₂ = N ₂ H ₂ + H ₂	1.74E+06	1.35	9.67E+03
NH ₂ + NH ₂ = H ₂ NN + H ₂	6.91E+20	-2.51	2.67E+04
NH ₂ + NH ₂ = NH ₃ + NH	2.15E+09	0.19	-3.93E+03
NH + OH = N + H ₂ O	2.00E+19	-2.24	-3.78E+04
NH ₃ + H = NH ₂ + H ₂	2.84E+25	-2.58	4.15E+04
NH + OH = HNO + H	4.11E+11	0.77	6.56E+04
N ₂ H ₄ + NH ₂ = N ₂ H ₃ + NH ₃	4.70E+20	-1.84	2.00E+04

driven pathways, which leads to excessive heat release and high flame speed. The Mei mechanism provides predictions that are reasonably accurate near $\phi \approx 1.1$, but it tends to overpredict LBVs under both $\phi < 0.85$ and stoichiometric conditions.

As pressure increases to 3 atm and 5 atm, a clear suppression of the LBV is observed, which agrees with both the simulation results and experimental measurements by Hayakawa [15] and Alvarez [73]. The peak LBV decreases from roughly 7.38 cm/s at 1 atm to 5.85 cm/s and 5.16 cm/s at 3 atm and 5 atm, respectively, verifying the retarding influence of high pressure on flame propagation. This pressure dependence is mainly attributed to two factors: the enhanced third-body collision effect at high pressure, which promotes chain termination reactions and slows the propagation of free radicals, and the reduced thermal diffusivity, which decreases flame thickness and propagation speed.

Additionally, experimental measurements at various pressures highlight the complexity of ammonia kinetics, arising from its relatively low reactivity, significant flame stretch sensitivity, and high ignition temperatures. Nevertheless, across the three pressure levels and varying equivalence ratios, the present model shows superior consistency with experimental data and outperforms many other mechanisms, thereby

offering reliable predictive capability for low-carbon ammonia combustion applications.

3.2.2. Laminar burning velocity for NH₃/O₂/N₂ mixture under various pressure condition

Fig. 8 shows the laminar burning velocities of ammonia in O₂ enrichment conditions as a function of equivalence ratio at pressures of 1, 2, and 5 atm. Here, Ω is defined as $\Omega = X_{O_2} / (X_{O_2} + X_{N_2})$. The lines denote the calculated results from this study and previous studies, while the symbols represent experimental data reported by Mei [48] and Yang [74].

Compared to NH₃/air flames, LBV values under oxygen enrichment conditions are significantly higher. At $\Omega = 0.35$ and $\phi \approx 1.1$ under 1 atm, the LBV increased from 7.38 cm/s in air to 25.83 cm/s. This significant improvement can be attributed to the higher oxygen content, which accelerates the propagation of NH₃. Despite the overall increase in flame speed, the peak LBV consistently occurred near $\phi \approx 1.1$ across all conditions, indicating that the fundamental combustion characteristics of ammonia are preserved.

As pressure increases from 1 atm to 5 atm at $\Omega = 0.35$, a consistent decrease in LBV is observed, which is consistent with trends observed in NH₃-air flames. However, the magnitude of pressure-induced suppression is notably smaller under oxygen-enriched conditions than in air. This suggests that oxygen enrichment mitigated the adverse effects of elevated pressure. Results from this study are consistent with experiments conducted under diverse conditions. The mechanism proposed by Song et al. overestimates the LBV under lean conditions, while the Otomo mechanism underestimates the LBV under rich conditions. The mechanism proposed by Mei et al. accurately predicts the flame speed but slightly overestimates the maximum flame speed. The Chen model overestimates LBV under all simulated conditions. For further analysis of LBV at additional oxygen enrichment levels, see the Supplementary Material S3.

3.3. Ignition delay time analysis

Fig. 9 presents the ignition delay time (IDT) of NH₃/Ar/O₂ mixture under various conditions, comparing the predictions of the present mechanism (solid lines) with those from the previous mechanisms [48, 66, 67] (dashed lines) and experimental data [16]. In the simulations, the ignition delay time is defined as the point at which a tangent drawn along the steepest gradient of the OH* de-excitation curve intersects the line corresponding to zero concentration.

In general, the proposed mechanism successfully reproduces the temperature dependence of the ignition delay within the studied range. The IDT decreases with increasing temperature, aligning well with the experimental observations. At both medium and high pressures (11 atm and 30 atm), the predicted IDTs closely match the measured values across the entire temperature span, with most discrepancies falling

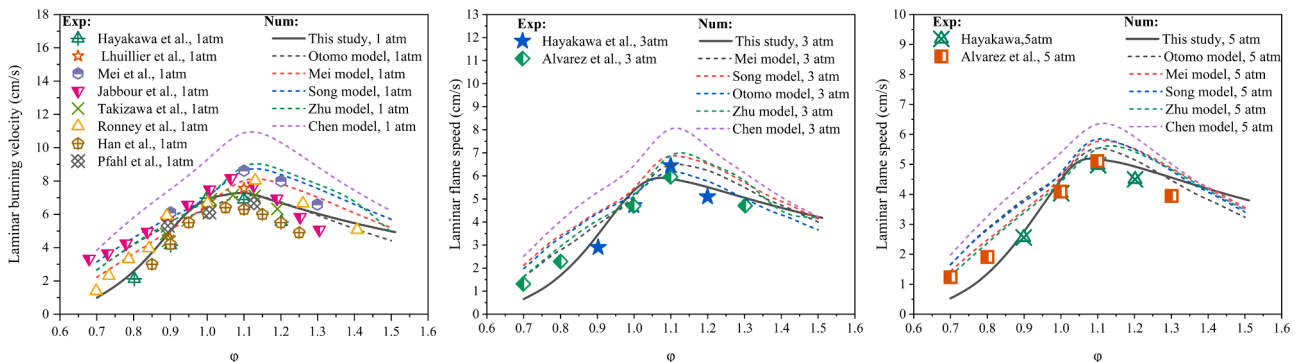


Fig. 7. Variation of laminar flame speed at $T = 300$ K with equivalence ratio under 1, 3, 5 atm. Solid line: This study; dashed lines: previous models [40,48,49,66, 67]; Scatter plot: experimental data from previous studies [12,13,48,68–73].

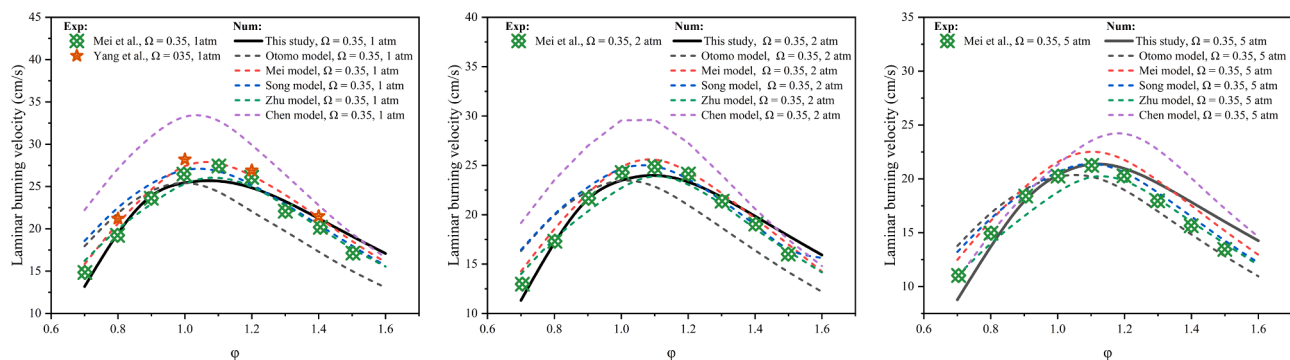


Fig. 8. Laminar burning velocity results of $\text{NH}_3/\text{O}_2/\text{N}_2$ mixtures at $T = 300 \text{ K}$, $\Omega = 0.35$ under 1, 2, 5 atm. Solid line: This study; dashed lines: previous models [40, 48, 49, 66, 67]; Scatter plot: experimental data from previous studies: [48, 74].

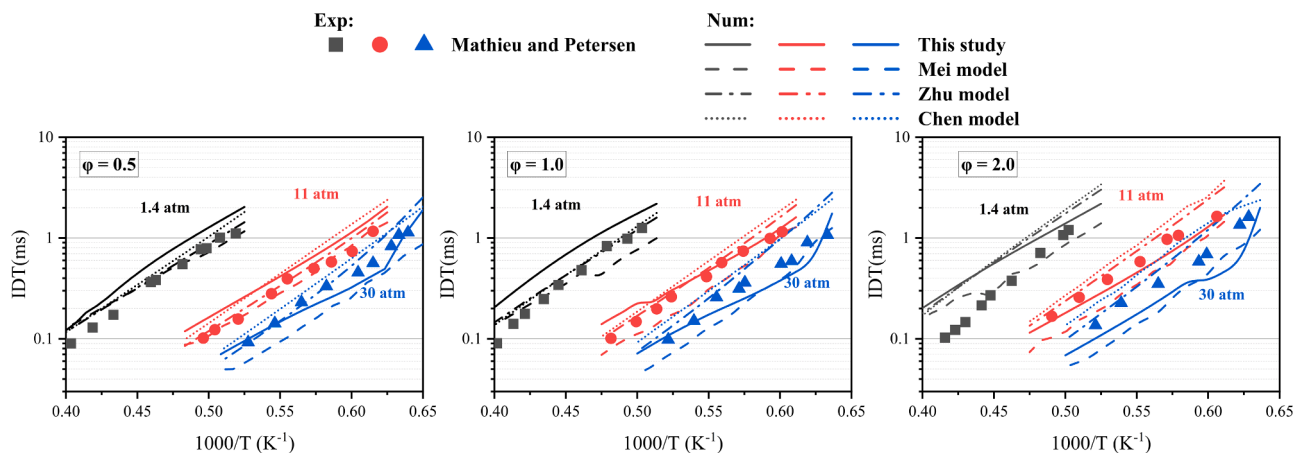


Fig. 9. Ignition delay time of $\text{NH}_3/\text{Ar}/\text{O}_2$ mixture with various equivalence ratios and pressures. Solid line: this study; dashed line: previous models [48, 66, 67]; scatter plot: experimental data from Mathieu and Petersen [16].

within the bounds of experimental uncertainty. However, this study overpredicts the IDT, particularly in the low-pressure region. This deviation is more pronounced than that observed with the Mei mechanism and yields slightly shorter delay times at 1.4 atm. However, that Mei mechanism tends to overestimate mixture reactivity under elevated pressures, resulting in a shortened IDT at medium and high pressures. Besides, the Zhu and Chen mechanisms generally underestimate mixture reactivity, leading to longer predicted IDT, particularly under elevated pressures.

The overestimation of IDT by the proposed mechanism under low-pressure conditions is likely attributed to limitations in the representation of pressure-dependent kinetics within the reaction mechanism. At lower pressure, the effects of low-pressure fall-off in radical recombination processes, as well as the competitive dynamics between chain-branching and chain-termination pathways, become significantly more

pronounced. These factors introduce additional complexity to NH_3 ignition behavior, particularly under low-temperature and low-pressure conditions. Further comparative calculations of IDT based on the peak concentrations of NH_2 radicals are provided in Supplementary Material S4.

3.4. Flame characteristic analysis

Fig. 10 presents the variation of adiabatic flame temperature and major combustion products, including H_2 , NH_3 , and NO , in NH_3/air premixed flames as a function of equivalence ratio at 1, 3, and 5 atm. The calculations are performed at an initial temperature of 300 K, and the calculation settings can be found in Section 2.2. Among the major combustion products, the mole fraction of hydrogen shows a strong correlation with the flame temperature. As a significant intermediate

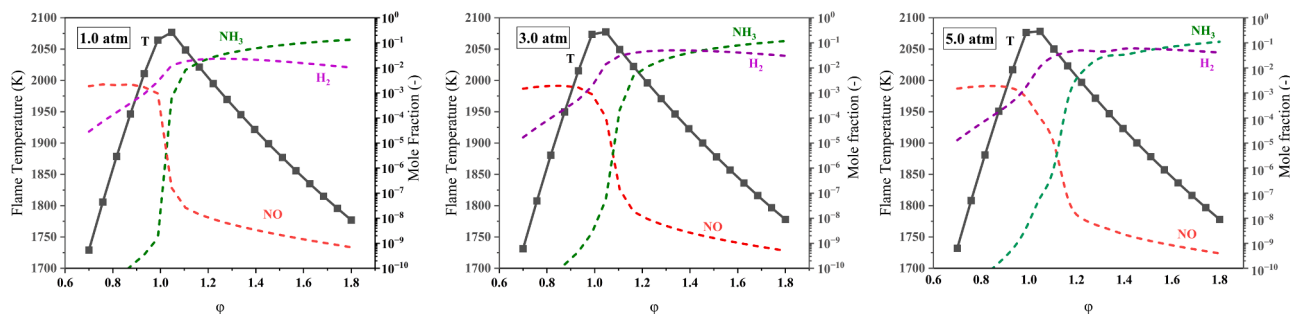


Fig. 10. Effect of equivalence ratio and pressure on adiabatic flame temperature and major combustion products in NH_3/air premixed flames.

species and energy carrier in ammonia combustion, hydrogen is primarily generated through the thermal decomposition of ammonia. In the lean regime, H_2 concentration increases with increasing equivalence ratio, reaching a peak in the vicinity of stoichiometric conditions, where fuel decomposition and radical formation are most intensive. NO remains at a high level, approximately 0.002 as O/H is abundant [4]. NO mole fraction begins to decrease as $\varphi > 1$. This trend is primarily attributed to the reduced availability of oxygen radicals, which limited the effectiveness of the NO production mechanism.

Increasing the pressure from 1.0 to 5.0 atm has minimal influence on the adiabatic flame temperature across the full range of equivalence ratios. The peak temperature remains nearly constant around $\varphi \approx 1.05$, and the adiabatic flame temperature is approximately 2080 K under all pressure conditions, indicating that the combustion heat release is primarily governed by thermodynamics rather than pressure-sensitive kinetics under idealized, well-mixed conditions.

Fig. 11 illustrates the distribution of key species in NH_3 /air flame profiles for $\varphi = 0.8, 1.0$, and 1.2 . The profiles depicted both the preheat zone and the reaction zone. The calculations are performed at an initial temperature of 300 K, and the calculation settings can be found in Section 2.2.

In all three cases, the reactant NH_3 is progressively consumed as the flame propagates, with the most rapid depletion occurring near the flame front, corresponding to the primary reaction zone. The consumption of NH_3 is accompanied by the rapid formation of intermediate radicals, particularly NH_2 , which emerges sharply within a narrow zone and then decays quickly downstream. NH_2 is a key intermediate in ammonia oxidation and serves as a branching point toward complete oxidation, producing H_2 and H_2O and nitrogen-containing species such as H_2NO and HNO. Under elevated pressure and intermediate temperature conditions, the downstream reactions of NH_2 proceed via H_2NO and HNO toward NO or N_2O production, with HO_2 acting as a principal chain carrier facilitating radical regeneration [49]. At $\varphi = 0.8$, H_2 reaches a modest mole fraction ($\sim 10^{-4}$), while at $\varphi = 1.2$, it accumulates to values exceeding 10^{-2} . This behavior reflects the pronounced influence of equivalence ratio on laminar flame speed. The laminar flame propagation velocity attains its maximum near $\varphi \approx 1.1$, thereby enhancing radical generation and intermediate consumption rates; deviations toward lean or rich conditions reduce the flame speed and concomitant radical pool, modifying downstream species profiles. The OH radical, a key indicator of flame reactivity and chain branching, forms sharply in the reaction zone and peaks near the flame front. The peak OH concentration remains relatively consistent across equivalence ratios, reflecting its pivotal role in maintaining chain reactions. As φ increases from 0.8 to 1.0, the peak NO concentration increases, indicating enhanced formation driven by high flame temperature, abundant radical availability, and effective branching via HNO. However, at $\varphi = 1.2$, the NO peak decreases noticeably. This reduction is attributable to the limited oxygen availability under fuel-rich conditions and a lower flame speed, leading to reduced OH and H radicals and therefore diminishing conversion of HNO to NO [75].

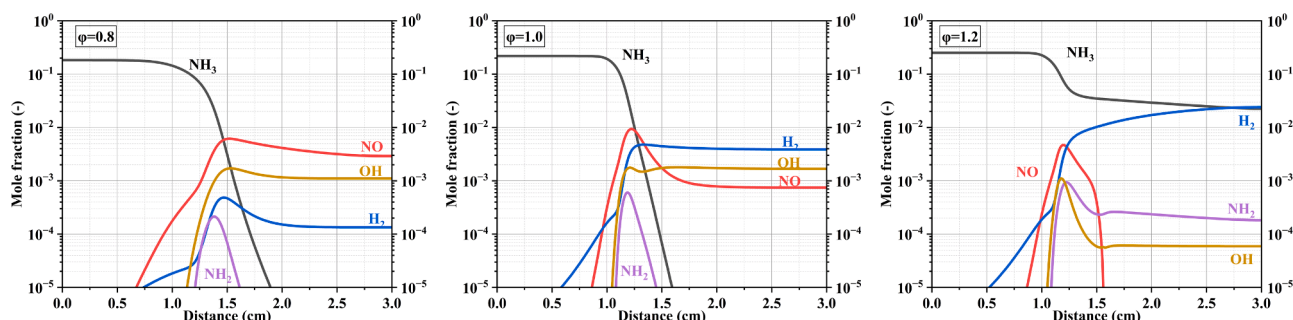


Fig. 11. Flame profiles of major species in NH_3 /air premixed flames at different equivalence ratios at 1 atm.

Fig. 12 represents the mole fraction of NO and N_2O under the jet-stirred reactor (JSR) oxidation results of NH_3 /air at equivalence ratios of 0.25 and 1.0, with a residence time $\tau = 1$ s, lines denote simulation results from different mechanisms, and the symbols present experimental data from Zhang et al. [76]. The updated mechanism reproduces the measured NO and N_2O profiles well at $\varphi = 0.25$, closely aligning the experimental trends, whereas the Song and Mei mechanisms markedly overpredict NO formation. Under lean conditions, the updated mechanism also depicts the low and relatively flat variations of NO and N_2O with temperature, avoiding the premature N_2O peak observed in previous mechanisms. However, the updated mechanism overpredicts the absolute concentrations of NO and N_2O at $\varphi = 1.0$. This discrepancy likely arises from differences in the underlying reaction pathways between lean and stoichiometric conditions. Since HNO generally dissociate or oxidize into NO, the NO deviation is primarily associated with the HNO-related kinetics and branching in NH_x -NO pathways [49,35]. For N_2O , the deviation mainly originates from differences in NH and NH_2 sub-mechanism, particularly through the $NH + NO = N_2O + H$ and $NH_2 + NO_2 = N_2O + H_2O$ channels that govern N_2O formation [77]. At $\varphi=1$, the relative concentration of NH_x intermediates differ significantly from the lean condition, thus resulting in a shift of the dominant reaction pathways. In particular, HNO-related branching reactions and NH_x -NO reaction channels assume a more significant role to local species concentrations. Consequently, the differences among kinetic mechanisms in the treatment of HNO decomposition and oxidation, NH_x reaction rate constants can be substantially amplified under stoichiometric conditions, thereby contributing to the observed overprediction of NO and N_2O .

Fig. 13 presents the rate of production (ROP) profiles of NO in NH_3 /air premixed flames at equivalence ratios $\varphi = 0.8, 1.0$, and 1.2 . The calculations are performed at an initial temperature of 300 K, calculation settings can be found in Section 2.2. The figure includes contributions from key elementary reactions to both NO formation and consumption as a function of flame distance. Fuel-N pathways dominate the NO concentration in ammonia/air combustion process [4,68], particularly through the HNO intermediate through the reaction $HNO + O_2 = HO_2 + NO$ [78,79].

At $\varphi = 0.8$, NO formation is primarily driven by $HNO + O_2 = HO_2 + NO$, emphasizing the importance of oxidative reactions under excess oxygen. The net NO ROP oscillates between positive and negative, showing a dynamic formation-consumption balance. At $\varphi = 1.0$, the flame exhibited the highest total NO ROP peak, centered sharply around 1.2 cm, with a maximum NO ROP was 6.60×10^{-5} (mol/cm³/s), coinciding with the main heat release zone. Formation reactions such as $NH + O_2 = NO + OH$ and $H + NO_2 = NO + OH$ intensified, which is attributed to increased concentrations of fuel-nitrogen intermediates. Consumption pathway $NH_2 + NO = H_2O + N_2$ remained dominant but insufficient to counter overall NO production. As φ increased to 1.2, the total NO ROP curve became flatter and of lower magnitude, with a peak NO ROP of 4.61×10^{-5} (mol/cm³/s), indicating suppressed NO production. NO consumption via $NH_2 + NO = H_2O + N_2$ became more

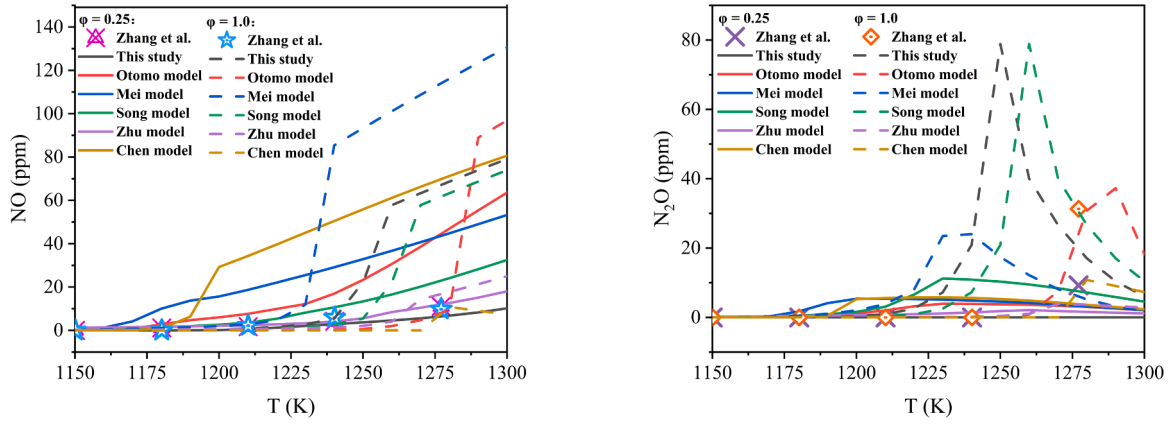


Fig. 12. Mole fraction of NO and N₂O in JSR oxidation of NH₃ at $\phi = 0.25$ and 1, $\tau = 1$ s, $P = 1$ atm. Solid and dashed lines: simulation results from previous mechanisms [40,48,49,66,67]; symbols: experimental data collected from Zhang et al. [76].

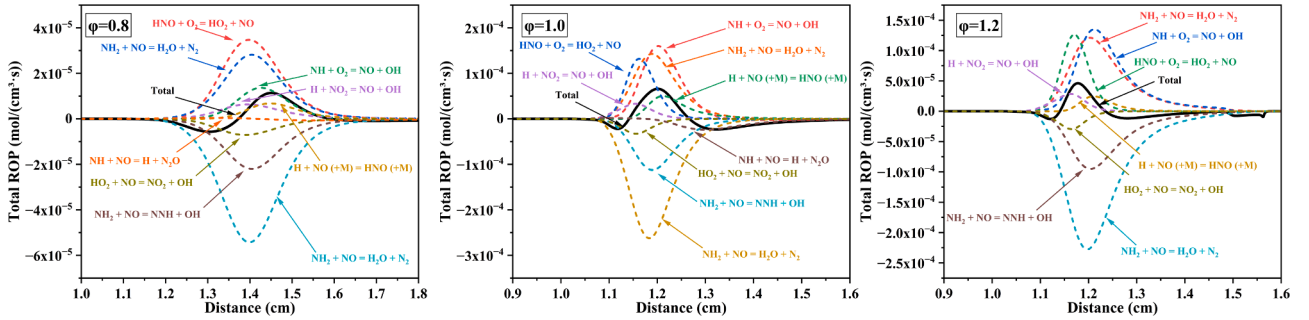


Fig. 13. Rate of production (ROP) profiles of NO in NH₃/air premixed flames at various equivalence ratios at 1 atm.

prominent, nearly balancing formation rates.

Additionally, the NO production peak shifts upstream with increasing ϕ , moving closer to the flame front. Under rich conditions, the consumption of NH₃ and the formation of reactive intermediates NH₂ and NH occur earlier due to higher fuel concentrations and limited oxygen diffusion, altering the flame structure and reaction zone position.

3.5. Temperature dependence of premix flame

The temperature dependence of laminar flame speed is a critical parameter in characterizing flame response to thermal conditions. The relationship is commonly described by the semi-empirical power-law correlation [72]:

$$\frac{S_L}{S_L^0} = \left(\frac{T_i}{T_i^0} \right)^{\alpha_T} \quad (1)$$

Where S_L and S_L^0 represent the laminar flame velocity at initial temperature T_i and reference temperature T_i^0 , respectively. The exponent α_T , known as the temperature dependence coefficient, quantifies the sensitivity of the flame speed to changes in the initial temperature and thus provides a quantitative measure of the influence of thermal conditions on flame propagation.

Fig. 13(a) presents the variation of α_T with equivalence ratio ϕ for NH₃/air premixed flame with $T_0 = 298$ K and $T_i = 498$ K, and experimental data from Zhou et al. [11] are included for comparison. Simulation results aligned well with the experiments, showing a non-linear trend where α_T reaches a minimum near $\phi \approx 1.05$, and increases on both lean and rich sides. Near stoichiometric conditions, optimal reactivity led to reduced temperature sensitivity, while under lean and rich conditions, lower reactivity and diffusion limitations, respectively, enhanced the temperature dependence.

Fig. 14(b) quantifies the temperature dependence of NH₃/N₂/O₂

premix flame with different oxygen contents at $\phi = 1$. The slope of the linear fits α_T decreases as oxidizer mole fraction Ω increases from 0.3 to 0.4, indicating reduced temperature sensitivity. Higher α_T reflects a stronger thermal activation of the flame, while lower values suggest kinetically dominated behavior. Thus, preheating significantly boosts flame speed in low-oxygen mixtures, but its effect diminishes under high-O₂ conditions.

4. Conclusions

In this work, a detailed kinetic mechanism with 33 species and 223 reactions for ammonia combustion is developed by updating key elementary reactions based on DFT calculations. The performance of the updated mechanism is systematically evaluated through flame speed, ignition delay time, and species concentration predictions. The main conclusions are as follows:

1. The updated kinetic mechanism demonstrates high predictive accuracy for the laminar flame speeds of NH₃/air mixtures across a wide range of equivalence ratios and pressures. The mechanism also reproduces experimental results under varying oxygen concentrations, confirming its applicability across diverse combustion conditions.
2. This mechanism accurately predicts the formation of NO and N₂O in jet-stirred reactor (JSR) experiments. ROP analysis performed in premixed flame indicates that NO formation is primarily controlled by the fuel-N pathway, specifically the reaction $\text{HNO} + \text{O}_2 = \text{HO}_2 + \text{NO}$. As ϕ increases, the NO formation rate peak shifts upstream, indicating earlier intermediate formation and a more compact reaction zone.
3. The temperature dependence coefficient α_T of NH₃/O₂/N₂ mixtures exhibits a nonlinear dependence on equivalence ratio. This coefficient reaches a minimum near stoichiometric conditions, suggesting

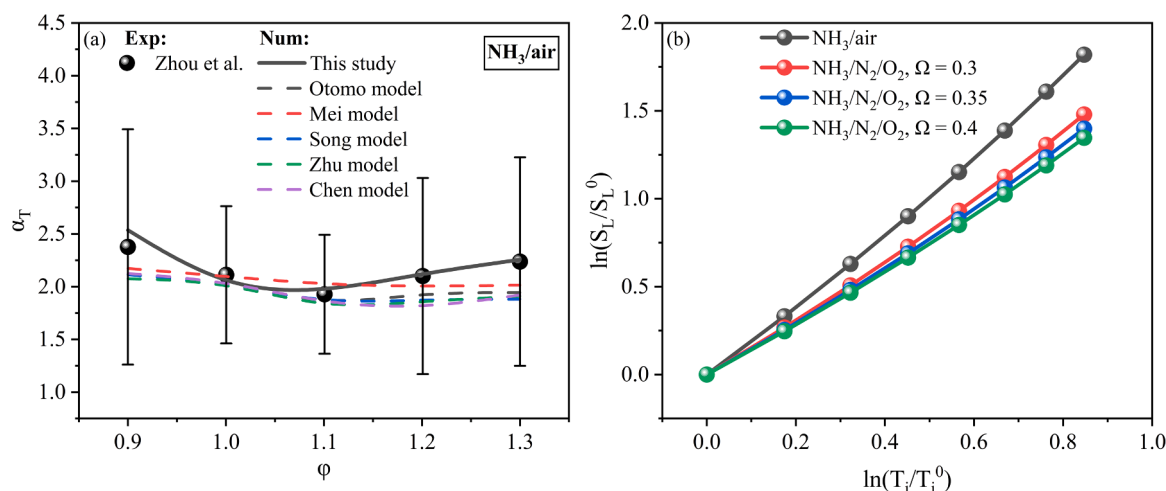


Fig. 14. Temperature dependence of LBV for $\text{NH}_3/\text{O}_2/\text{N}_2$ mixtures. (a) Variation of the temperature dependence coefficient α_T with equivalence ratio ϕ . Black solid lines: this study; dashed lines: previous models [40,48,49,66,67]; scatter plot: experimental data from Zhou et al. [11]. (b) Temperature dependence of $\text{NH}_3/\text{O}_2/\text{N}_2$ with different oxygen contents.

a lower sensitivity of reaction rate to temperature. at optimal fuel–oxidizer ratios. In $\text{NH}_3/\text{O}_2/\text{N}_2$ mixtures, α_T decreased with increasing oxygen concentration, indicating a transition from a thermally controlled regime to a kinetically dominated one.

CRediT authorship contribution statement

Ke Ren: Writing – original draft, Methodology, Investigation, Formal analysis. **Xiaoping Chen:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis. **Jiliang Ma:** Writing – review & editing, Supervision. **Suyang Pan:** Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2025.114697](https://doi.org/10.1016/j.combustflame.2025.114697).

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